

Trial Test 9

- Submit your answers via Google Forms (you have only 1 hour and 10 mins to complete).
- Your score will be available immediately after the test. Please, check your emails.
- If you have technical issues, let me know in DM or in the Group Chat.
- You are not allowed to use the calculator. Also, do not google, be honest!
- Do not panic and manage your time wisely. Read the questions carefully and try to answer all of them.

Good Luck!

1) Evaluate

$$(9a^2 + \sqrt{a})(a^2 + \sqrt{a})$$

a) $9a^4 + 10a^2 * \sqrt{a} + a$

b) $9a^3 + 10a^2 + \sqrt{a}$

c) $9a^4 + 10a^{2.6} + a$

d) $9a^4 + 10a^2 + \sqrt{a}$

2) Find the sum of the first 15 terms of an arithmetic sequence, if $a_8 = 16$

a) 240

b) 280

c) 300

d) 360

3) Find the value of product of roots:

$$4x^2 - \frac{25}{7}x + 38 = 0$$

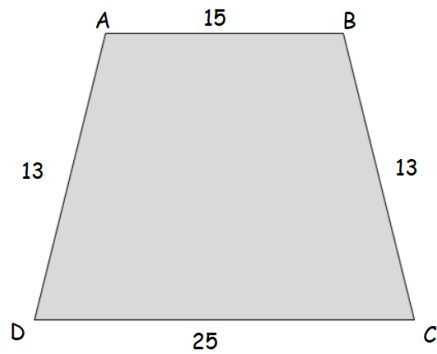
a) $\frac{-25}{28}$

b) $\frac{25}{28}$

c) $\frac{19}{2}$

d) $\frac{-19}{2}$

4) Find the area of trapezoid, if $AB=15$, $CD=25$ and $AD=BC=13$



(the picture is not to scale)

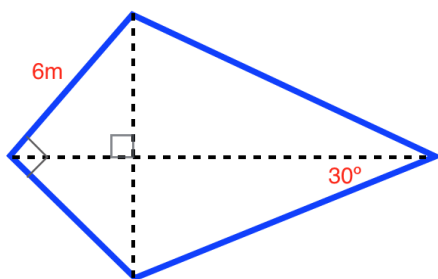
a) 220

b) 240

c) 260

d) 280

5) Find the area of the given kite



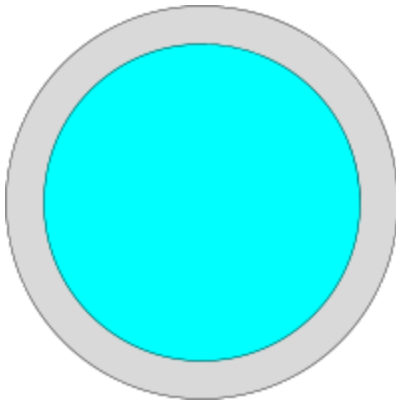
a) $16(\sqrt{2} + 1)$

b) $16(\sqrt{3} + 1)$

c) $18(\sqrt{2} + 1)$

d) $18(\sqrt{3} + 1)$

6) Find the radius of the large circle, if the radius of the small circle is less than the large one by 2 meters, and ratio of the area of smaller circle to area of big circle is 81:100



a) 18

b) 20

c) 22

d) 24

7) Given that the volume of the planet core is $\frac{1}{10}$ of the volume of the entire planet.

Find the ratio of $\frac{r}{R}$, where r - radius of planet core, R - radius of planet

a) $\frac{\sqrt[2]{10}}{10^2}$

b) $\frac{\sqrt[3]{10^2}}{10}$

$$\text{c) } \frac{\sqrt[3]{10}}{10}$$

$$\text{d) } \frac{\sqrt[2]{10^3}}{10^2}$$

8) Rearrange the formula in terms of Z:

$$X = \sqrt{17 - \sqrt{2(Z + 7)^2 - 9}}$$

$$\text{a) } Z = \sqrt{\frac{(17 - X^2) + 9}{2}} - 7$$

$$\text{b) } Z = \sqrt{\frac{(X + 17)^2 + 9}{2}} - 7$$

$$\text{c) } Z = \sqrt{\frac{(X^2 + 17)^2 - 9}{2}} + 7$$

$$\text{d) } Z = \sqrt{\frac{(17 - X)^2 - 9}{2}} + 7$$

9) B is 15km away from A on the bearing of 60° , C is 15km away from B on a bearing of 120° . What

is a distance between A and C

$$\text{a) } 15$$

$$\text{b) } 15\sqrt{3}$$

c) $15\sqrt{2}$

d) $15\sqrt{6}$

10) Find the equation of the line passing through the point P(3, -2) which has positive intercepts on the x- and y-axis in the ratio 4:3

a) $y = -\frac{3}{4}x + \frac{1}{4}$

b) $y = -\frac{1}{4}x + \frac{3}{4}$

c) $y = \frac{3}{4}x + \frac{1}{4}$

d) $y = \frac{1}{4}x + \frac{3}{4}$

11) Converts between 37812 cm^3 to litres

a) 37,812

b) 3,7812

c) 378,12

d) $37,812 * 10^{-4}$

12) Find the volume of frustum cone, if the $R_1 = 15$, $r_2 = 5$, and *slant height* = $10\sqrt{2}$

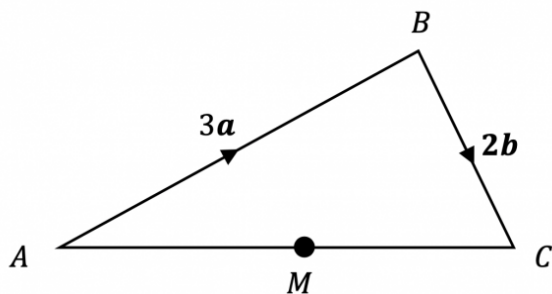
a) $\frac{3250\pi}{3}$

b) $\frac{2500\pi}{3}$

c) $\frac{325\pi}{3}$

d) $\frac{250\pi}{3}$

13) Find an expression of vector $\vec{AM}$ in terms of $\vec{a}$ and $\vec{b}$



a) $\frac{3\vec{a}}{2} + \vec{b}$

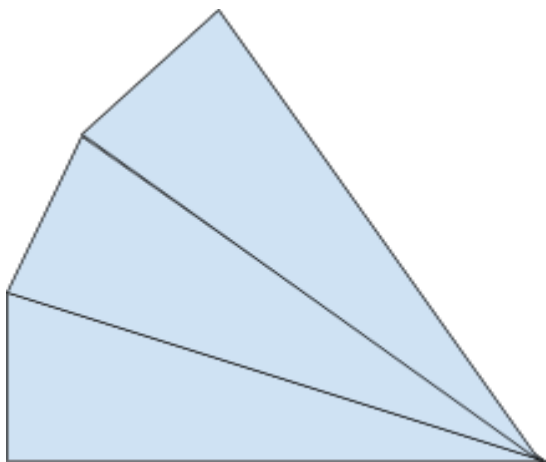
b) $-\frac{3\vec{a}}{2} + \vec{b}$

c) $-\frac{3\vec{a}}{2} - \vec{b}$

d) $\frac{3\vec{a}}{2} - \vec{b}$

14) The diagram shows three similar right-angled triangles, the first(the smallest) has legs 2cm and 5 cm in length.

Find the area of third triangle



a) $\frac{841}{125}$

b) $\frac{751}{125}$

c) $\frac{841}{25}$

d) $\frac{751}{25}$

15) Find the value of an exterior angle of regular polygon with 12 sides

a) 15

b) 30

c) 45

d) 60

16) Evaluate and write in standard form

$$\frac{2 \cdot 10^9 + 8 \cdot 10^6 - 16 \cdot 10^{-9}}{16 \cdot 10^{-5}}$$

a) $12.55 \cdot 10^{12}$

b) $1.255 \cdot 10^{13}$

c) $125.5 \cdot 10^{11}$

d) $0.1255 \cdot 10^{14}$

17) First worker finishes the work in 3 hours, while the second worker spends 8 hours finishing the same task.

How many hours will it take to finish the task together?

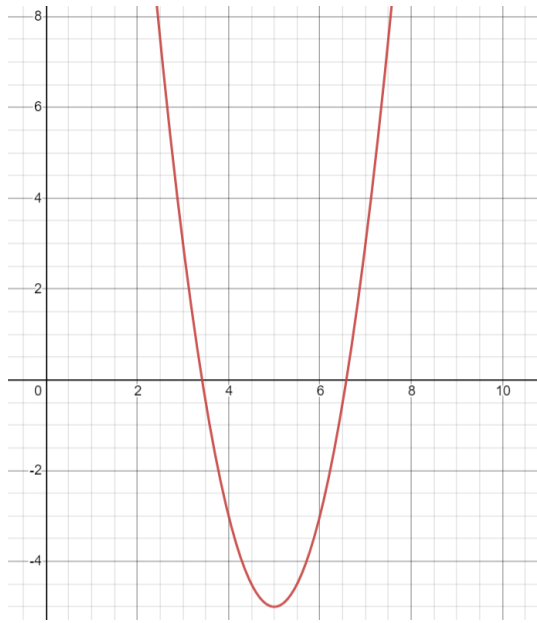
a) 11 h

b) 5 h

c) $\frac{24}{11} \text{ h}$

d) $\frac{24}{5} \text{ h}$

18) Find the equation of the parabola:



a) $2(x + 5)^2 + 5$

b) $2(x - 5)^2 + 5$

c) $2(x + 5)^2 - 5$

d) $2(x - 5)^2 - 5$

19) Convert the decimal to fraction:

$0.(54) + 0.(576)$

a) $\frac{4}{11} + \frac{84}{111}$

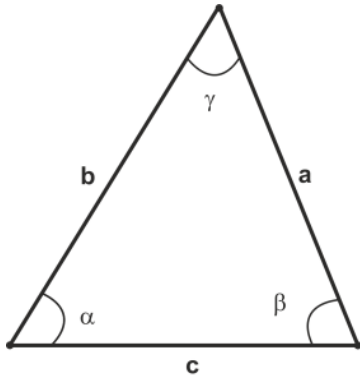
b) $\frac{7}{11} + \frac{32}{111}$

c) $\frac{6}{11} + \frac{64}{111}$

d) $\frac{9}{11} + \frac{48}{111}$

20) Given that angle α is equal to 60 degrees, and sides $b = 15$ $c = 12$.

Find a .



(figure is not to scale)

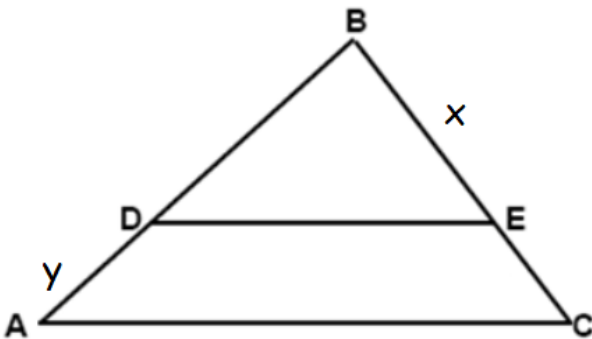
a) $3\sqrt{31}$

b) $3\sqrt{19}$

c) $3\sqrt{15}$

d) $3\sqrt{13}$

21) Find the values of x and y , if given that $DE = 6$, $AC = 8$, $BC = 12$, $DB = 12$



(figure is not to scale)

a) $x = 9$, $y = 4$

b) $x = 8, y = 6$

c) $x = 6, y = 8$

d) $x = 4, y = 9$

22) $a \bullet b = a^2 - b^2, a \diamond b = \frac{1}{2}(a \bullet b).$

Find the value of $(1 \bullet 4) \bullet (4 \diamond 6)$

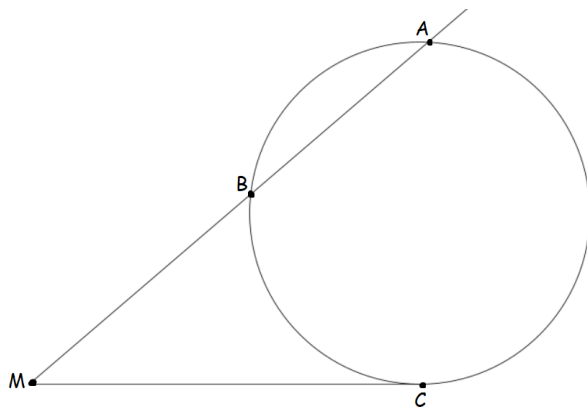
a) 75

b) 125

c) 225

d) 325

23) Find the value of MC, if given that MB = 7 and AB = 5



a) $2\sqrt{23}$

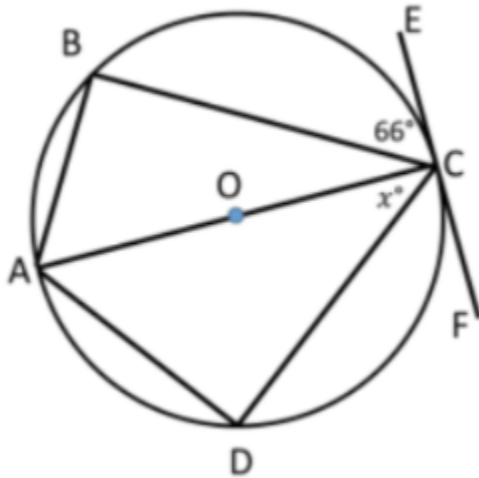
b) $2\sqrt{21}$

c) $2\sqrt{19}$

d) $2\sqrt{17}$

24) Given that EF is tangent to the circle, and the angle BAD has a value of $2x$.

Find the value of x



a) 46

b) 52

c) 48

d) 64

25) Find the value of x , if:

$$4^x * 16^y = 256^4$$

$$\frac{1}{3^y} = 243$$

a) 18

b) 21

c) 24

d) 26

26) Solve inequality

$$\frac{(x-5)(x^2-5x+6)(x-9)}{x+2} \leq 0$$

a) $(-\infty, -2) \cup (2, 3] \cup [5, 9]$

b) $(-\infty, 2) \cup (2, 4) \cup (-5, 9)$

c) $(-\infty, -2) \cup [2, 3] \cup [5, 9]$

d) $(-\infty, 2) \cup [2, 4] \cup [-5, 9]$

27) If the time on the clock is 7:50.

What is the value of the angle between hour and minute hands?

a) 55

b) 65

c) 75

d) 85

28) Find the sum of the roots of the quadratic equation

$$\frac{8}{3}x^2 - 2x + 33 = 0$$

a) $-\frac{3}{4}$

b) $\frac{3}{4}$

c) $-\frac{99}{8}$

d) $\frac{99}{8}$

29) A mixture of 80 liters of alcohol and water contains 25% water. How much water must be added to above mixture to make the water 80% of the resulting mixture

a) 200

b) 220

c) 240

d) 260

30) If an amount of P is compounded with 10% interest rate.

Show the final amount after 3 years in terms of P

a) 1.04P

b) 1,441P

c) 1,331P

d) 1.432P

как ты?

устал да?

я тоже

приляг отдохни



tg: @melonkid411

